Multi-Stage Area Probability Sample Design for the Detroit Transportation Security Survey

SURV 745 – PracTools Project Team Neyman Namit Shrivastava, Nicholas Reign Lugu | @ namit507@gmail.com

April 11, 2026

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1 Introduction

This report presents the sample design for a face-to-face survey of residents of the City of Detroit, Michigan. The survey examines transportation security and transportation use. Key variables include the proportion of residents living in a vehicle-free household, the proportion whose main means of transportation is public transit, the proportion who skip going somewhere due to transportation problems, and the proportion who asked friends, family, or neighbors for transportation help in the past month.

The design uses a multi-stage area probability sample in which census tracts serve as primary sampling units (PSUs), block groups (BGs) serve as secondary sampling units (SSUs), and persons are selected from area listings within sampled BGs at the final stage. The design meets a target of 1,000 responding adults, allocated equally across four age groups, while accounting for differential expected response rates. The remainder of this report describes the frame construction, selection procedures, anticipated precision, weighting plan, and variance estimation guidance.

2 Sample Design

2.1 Goals of the Sample Design

The goals of the sample design are to: (1) select a probability sample yielding approximately 1,000 responding adults from the City of Detroit, with 250 respondents in each of four age groups (18–34, 35–49, 50–64, and 65+); (2) select 40 census tracts and, within each selected tract, 1 block group; (3) maintain a self-weighting design within each age group so that every person in a given age domain has the same overall probability of selection; and (4) achieve approximately equal field workload across selected tracts by using probabilities proportional to a composite measure of size reflecting the required age-specific sampling rates.

Table 1: Target Completed Interviews and Expected Response Rates

Age Group	Target Respondents	Expected Response Rate
18-34	250	0.35
35-49	250	0.45
50-64	250	0.50
65+	250	0.65

2.2 Target Population

The target population consists of all civilian, non-institutionalized persons aged 18 years and older residing in the City of Detroit, Michigan. Persons living in institutional settings are excluded because the survey is designed as a household-based face-to-face survey.

2.3 Area Sampling Frame: Units, Data Available, and Source of the Data

2.3.1 Data Sources

The area sampling frame is constructed from two sources: (1) a file of Detroit census tract FIPS codes obtained from the City of Detroit Open Data Portal; and (2) a Detroit census data file containing tract-level and block-group-level counts from the 2020 Decennial Census Demographic and Housing Characteristics (DHC) Summary File, including total households, total persons, and age-by-sex counts from table P12.

2.3.2 Frame Construction

The initial tract file is deduplicated at the tract level because some special geographic units appear more than once in the source file. Tracts with zero adult population and tracts with fewer than 50 adults are removed from the operational frame; these units are not meaningful household survey PSUs and would create unstable sampling rates and unnecessary field burden. Tracts with only one block group are retained and that single block group is automatically selected at stage 2 (conditional probability = 1.0).

Table 2: Summary of Frame Construction

Step	Count
Initial tract records after deduplication	276
Tracts with zero adults aged 18+	6
Tracts with 1–49 adults aged 18+	4
Total tracts removed	10
Final tract frame	266
Final block group frame	614

2.4 Assigning Measures of Size to Units

A central requirement of the design is to achieve 250 completed interviews in each age group despite differing expected response rates. For age group g, the required person-level sampling fraction is:

$$f_g = \frac{n_g}{r_g N_g}$$

where n_g is the target number of completed interviews, r_g is the expected response rate, and N_g is the Detroit population total in age group g.

To support both the age-group targets and approximately equal field workload, the design uses the following composite measure of size (MOS) for each unit i:

$$MOS_i = \sum_{g=1}^{4} f_g \, N_{g,i}$$

where $N_{g,i}$ is the count of persons in age group g in unit i. This MOS reflects the expected pre-response workload implied by the differential sampling fractions required by each age domain.

Table 3: Age-Specific Sampling Fractions Used in the Composite MOS

Age Group	Population Total	Target Respondents	Expected Response Rate	Required Sampling Fraction
18–34	161344	250	0.35	0.004427
35–49	113270	250	0.45	0.004905
50–64	117755	250	0.50	0.004246
65+	87733	250	0.65	0.004384

The total composite MOS is 2,154.46. With 40 tracts selected, the design implies an expected pre-response workload of 53.86 listed adults per selected tract, versus an expected 25.0 completed interviews per tract.

The frame contains 266 operational tracts and 614 operational block groups. The tract-level composite MOS varies substantially across the city, supporting the use of PPS selection to spread the sample efficiently while preserving target age-group yields.

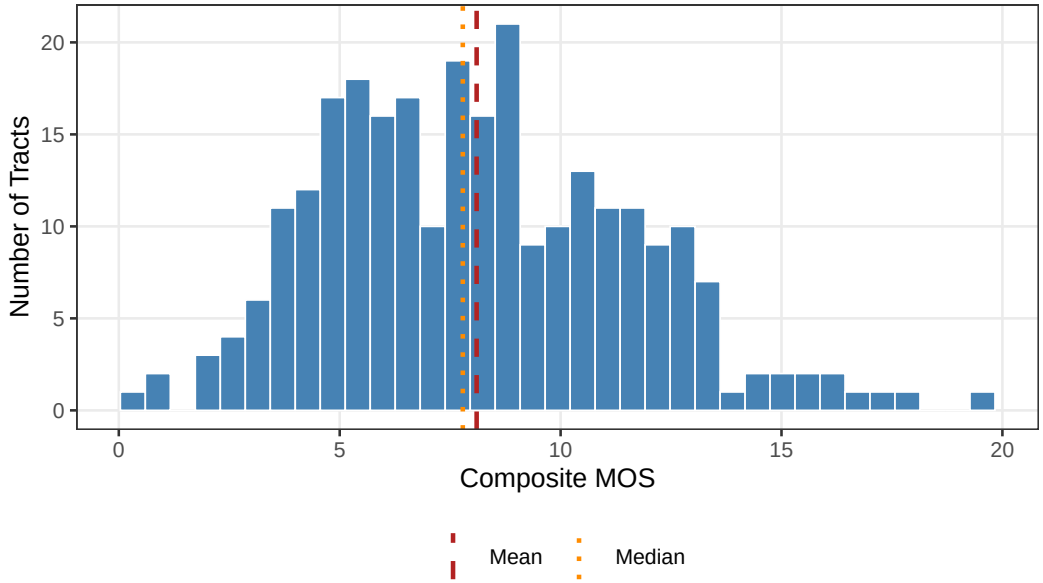

Figure 1: Distribution of the Composite MOS Across Detroit Census Tracts. The skewed distribution confirms that equal-probability selection would produce unequal workloads; PPS selection equalizes workload by giving larger tracts higher inclusion probabilities.

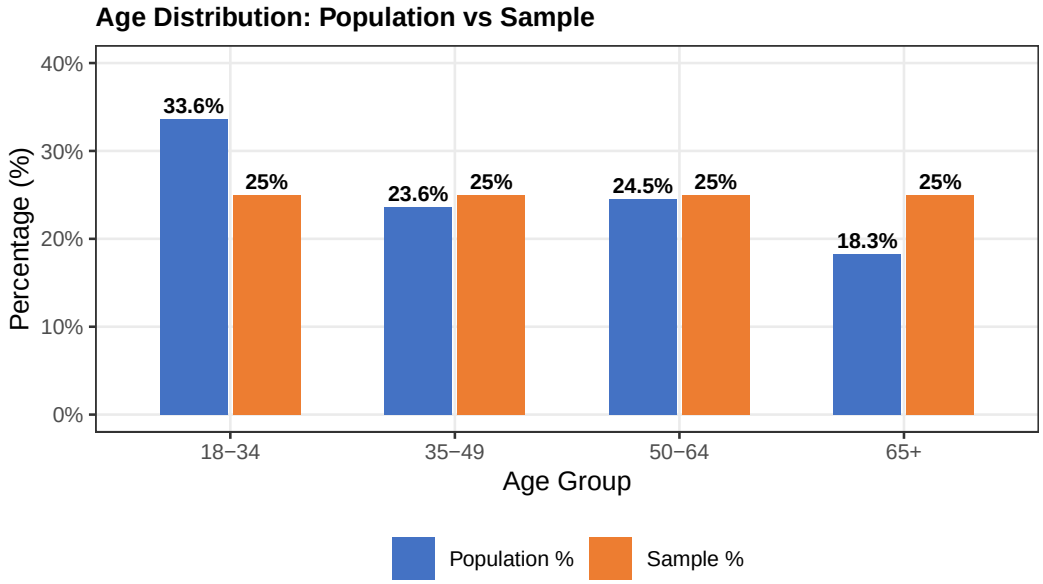

Figure 2: Age Distribution: Population Share vs. Equal Sample Allocation. Blue bars show each age group's share of the Detroit adult population; orange bars show the equal 25% allocation in the sample. The 18–34 group is overrepresented in the population relative to its sample share, while the 65+ group is underrepresented, motivating the use of domain-specific sampling fractions.

Table 4: Descriptive Statistics for the Detroit Tract Frame (N = 266 tracts)

Variable	Mean	Median	SD	Min	Max
Total Persons	2402.6	2247.5	1045.9	83.0	5518.0
Total Households	1165.0	1123.0	473.0	39.0	3906.0
Adults 18+	1804.9	1736.5	767.3	71.0	4417.0
Age 18–34	606.6	571.0	333.3	20.0	3241.0
Age 35–49	425.8	400.5	191.1	17.0	1084.0
Age 50–64	442.7	422.0	184.5	16.0	1168.0
Age 65+	329.8	291.0	194.4	6.0	1623.0
Composite MOS	8.1	7.8	3.4	0.3	19.5

Table 5: Distribution of the Number of Block Groups per Tract

Block Groups per Tract	Number of Tracts
1	40
2	134
3	68
4	19
5	4
6	1

2.5 Could a Single Sampling Rate Meet the Age-Group Targets?

A useful alternative is a design applying the same person-level sampling rate f to all adults. Under this design, f satisfies:

$$f = \frac{1000}{\sum_g N_g r_g}.$$

Table 6: Expected Respondents Under a Design with the Same Sampling Rate for All Persons

Age Group	Population	Exp. Response Rate	Target	Exp. Respondents	Difference
18–34	161344	0.35	250	252.8	2.8
35–49	113270	0.45	250	228.2	-21.8
50–64	117755	0.50	250	263.6	13.6
65+	87733	0.65	250	255.3	5.3

An equal-rate design would require a sampling rate of 0.004477. Its principal advantage is simplicity: it produces a fully self-weighting sample for all adults, reducing weighting variance. Its disadvantage is that it cannot meet the age-domain targets exactly. As the table and figure below show, the 35–49 group would fall materially short of 250, while the 50–64 and 65+ groups would overshoot. Because the four domain-specific targets are fixed requirements, the equal-rate design is not appropriate.

3 Sample Selection

3.1 Method of Selection

The sample is selected in three stages:

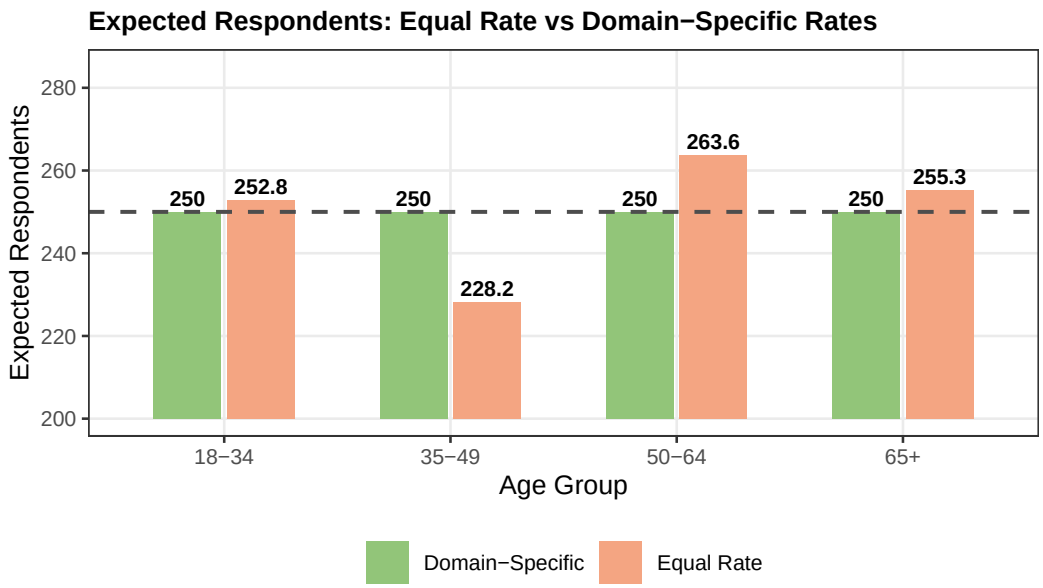

Figure 3: Expected Respondents Under Equal-Rate vs. Domain-Specific Sampling. The dashed line marks the 250-respondent target for each age group. The equal-rate design misses the target for the 35–49 group and overshoots for 50–64 and 65+; the domain-specific design hits 250 exactly in all groups.

1. **Stage 1 (PSUs):** Select 40 census tracts from the tract frame using systematic PPS sampling with the composite MOS, with the frame sorted in ascending FIPS-code order.
2. **Stage 2 (SSUs):** Within each selected tract, select exactly 1 block group using PPS sampling with the block-group-level composite MOS. For tracts containing only one block group, that block group is selected with certainty.
3. **Stage 3 (persons):** Within each sampled block group, list all occupied housing units and all adults aged 18+. Within each age domain, select persons at the domain-specific rate $f_{g,ij}$ required to yield the expected completed interviews.

3.2 Selected Units and Their Characteristics

3.3 Selection Probabilities at Each Stage

For tract i, the first-stage inclusion probability is computed using the inclusionprobabilities() function from the R sampling package, which rescales the raw PPS probabilities to sum to exactly $n_I = 40$ while correctly assigning probability 1 to any tract whose raw probability exceeds 1 (certainty units):

$$\pi_{1i} = \text{inclusionprobabilities}(MOS_i,\ 40), \quad \sum_i \pi_{1i} = 40.$$

For block group j within selected tract i, the conditional second-stage probability is:

$$\pi_{2j|i} = \frac{MOS_{ij}}{\sum_{\ell \in i} MOS_{i\ell}},$$

Table 7: Selected PSUs (Census Tracts) and Their Characteristics

Tract FIPS	Name	Total Pop	Adults 18+	Composite MOS	Inclusion Prob.	No. BGs
26163500200	Census Tract 5002	3037	2088	9.41	0.1746	2
26163500900	Census Tract 5009	3706	2573	11.58	0.2151	2
26163501400	Census Tract 5014	3624	2685	12.07	0.2241	3
26163502000	Census Tract 5020	1858	1289	5.81	0.1078	1
26163503400	Census Tract 5034	886	673	3.03	0.0562	1
26163505100	Census Tract 5051	2888	2253	10.08	0.1871	3
26163506100	Census Tract 5061	1875	1379	6.19	0.1149	1
26163506900	Census Tract 5069	3579	2551	11.47	0.2130	3
26163508000	Census Tract 5080	1098	835	3.74	0.0695	2
26163511400	Census Tract 5114	2005	1675	7.48	0.1388	3
26163513900	Census Tract 5139	1556	1165	5.20	0.0966	2
26163515700	Census Tract 5157	4618	4417	19.55	0.3630	3
26163516700	Census Tract 5167	3218	2604	11.61	0.2155	3
26163517500	Census Tract 5175	2856	2528	11.21	0.2080	2
26163520300	Census Tract 5203	2622	2505	11.19	0.2078	3
26163521800	Census Tract 5218	1466	1179	5.25	0.0976	1
26163523300	Census Tract 5233	2914	2023	9.15	0.1699	3
26163524200	Census Tract 5242	4366	2884	13.08	0.2429	3
26163525700	Census Tract 5257	4403	2906	13.20	0.2451	4
26163526400	Census Tract 5264	1417	964	4.35	0.0808	2
26163530800	Census Tract 5308	1394	1092	4.88	0.0906	2
26163531800	Census Tract 5318	1500	1127	5.04	0.0937	2
26163533600	Census Tract 5336	1390	1035	4.66	0.0864	2
26163534700	Census Tract 5347	3056	2214	9.92	0.1843	3
26163535600	Census Tract 5356	3727	2896	12.95	0.2404	4
26163536500	Census Tract 5365	2097	1597	7.18	0.1332	2
26163537200	Census Tract 5372	583	450	2.03	0.0377	1
26163538200	Census Tract 5382	1972	1624	7.26	0.1349	2
26163538600	Census Tract 5386	4758	3619	16.21	0.3010	4
26163539200	Census Tract 5392	4863	3908	17.48	0.3245	3
26163539600	Census Tract 5396	3197	2439	10.93	0.2030	2
26163540400	Census Tract 5404	2632	2026	9.07	0.1684	2
26163540900	Census Tract 5409	3420	2641	11.83	0.2197	3
26163541500	Census Tract 5415	3625	2897	12.95	0.2405	2
26163542300	Census Tract 5423	2255	1740	7.82	0.1452	2
26163543100	Census Tract 5431	2455	1915	8.57	0.1591	2
26163544000	Census Tract 5440	2534	1985	8.90	0.1652	1
26163545500	Census Tract 5455	3878	2590	11.69	0.2171	3
26163545900	Census Tract 5459	3758	2530	11.41	0.2118	2
26163546700	Census Tract 5467	2816	2091	9.38	0.1741	2

and the overall block-group probability is

$$\pi_{ij}^{BG} = \pi_{1i}\,\pi_{2j|i}$$

.

For a person in age group g in selected block group j, the within-BG sampling rate is $f_{g,ij} = f_g/\pi_{ij}^{BG}$, so the overall person probability is:

$$\pi_{g,ij} = \pi_{ij}^{BG} \cdot f_{g,ij} = f_g,$$

confirming that the design is self-weighting within each age group.

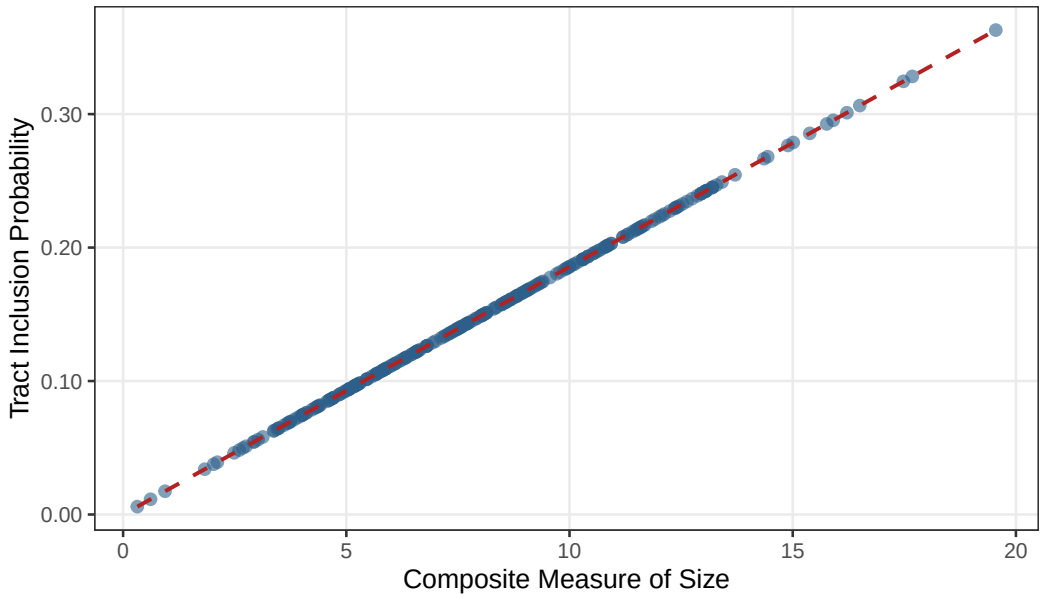

Figure 4: Tract Inclusion Probabilities vs. Composite Measure of Size. The near-perfect linear relationship confirms that systematic PPS selection correctly assigns each tract an inclusion probability proportional to its composite MOS.

Table 8: Summary of Selection Probabilities and Workload

Metric	Value
Number of selected tracts	40.0000
Number of selected block groups	40.0000
Mean tract inclusion probability	0.1740
Minimum tract inclusion probability	0.0377
Maximum tract inclusion probability	0.3630
Mean block-group overall inclusion probability	0.0803
Expected persons approached per sampled tract	53.8614
Expected completed interviews per sampled tract	25.1819
Operational persons selected per sampled tract	53.8750
Operational expected completed interviews per sampled tract	25.0087

3.4 How Persons Should Be Selected from Area Listings

For each sampled block group, interviewers should first prepare a complete listing of all occupied housing units and all adults aged 18+ within them, classifying each adult into one of the four age domains. Separate within-BG lists are then formed for each domain.

Within each domain, persons should be selected using the integer counts shown in the sample file. These counts are derived by a largest-remainder (Hamilton method) allocation: each BG's share of the domain total workload (`listed_g`) is first expressed as a proportional share of the domain target (e.g., 714 for age 18–34), then floored to integers, with any remaining units assigned to BGs with the largest fractional parts. This ensures the integer counts sum exactly to the domain target and are distributed proportionally to the BG-level workload. Field staff within each BG may use systematic sampling with a random start or random-number assignment within each age-domain list.

Table 9: Persons Selected and Expected Completed Interviews by Age Group

Age Group	Persons Selected	Expected Completed Interviews
18–34	714	249.9
35–49	556	250.2
50–64	500	250.0
65+	385	250.2
Total	2155	1000.4

4 Anticipated Precision

The key precision requirement concerns the estimated proportion of Detroit adults whose main means of transportation is public transit. Under a complex sample design:

$$Var(\hat{p}) = DEFF \times \frac{p(1-p)}{n},$$

where $n = 1{,}000$ and $DEFF$ is the design effect. Because the design has one sampled BG per tract, within-cluster homogeneity still creates a design effect greater than 1. The table below reports anticipated precision across plausible design effects (1.2 to 1.8) and several assumed proportions.

Table 10: Anticipated Precision for Estimated Proportions Under Plausible Design Effects

Assumed Proportion	DEFF	Standard Error	Margin of Error	95% CI
0	1.2	0	0.0204	[0.080, 0.120]
0	1.5	0	0.0228	[0.077, 0.123]
0	1.8	0	0.0249	[0.075, 0.125]
0	1.2	0	0.0242	[0.126, 0.174]
0	1.5	0	0.0271	[0.123, 0.177]
0	1.8	0	0.0297	[0.120, 0.180]
0	1.2	0	0.0272	[0.173, 0.227]
0	1.5	0	0.0304	[0.170, 0.230]
0	1.8	0	0.0333	[0.167, 0.233]
0	1.2	0	0.0294	[0.221, 0.279]
0	1.5	0	0.0329	[0.217, 0.283]
0	1.8	0	0.0360	[0.214, 0.286]
0	1.2	0	0.0311	[0.269, 0.331]
0	1.5	0	0.0348	[0.265, 0.335]
0	1.8	0	0.0381	[0.262, 0.338]

If the true public-transit proportion is around 0.15, the margin of error ranges from approximately ± 0.024 (DEFF = 1.2) to ± 0.030 (DEFF = 1.8), indicating adequate citywide precision. The equal domain allocation also supports reliable age-subgroup estimates.

5 Description of Weighting Plan

Because the design is self-weighting within each age group, the base weight for a respondent in age group g is constant within that group:

$$w_{\text{base},g} = \frac{1}{f_g}.$$

After data collection, a nonresponse adjustment should be applied within weighting cells defined

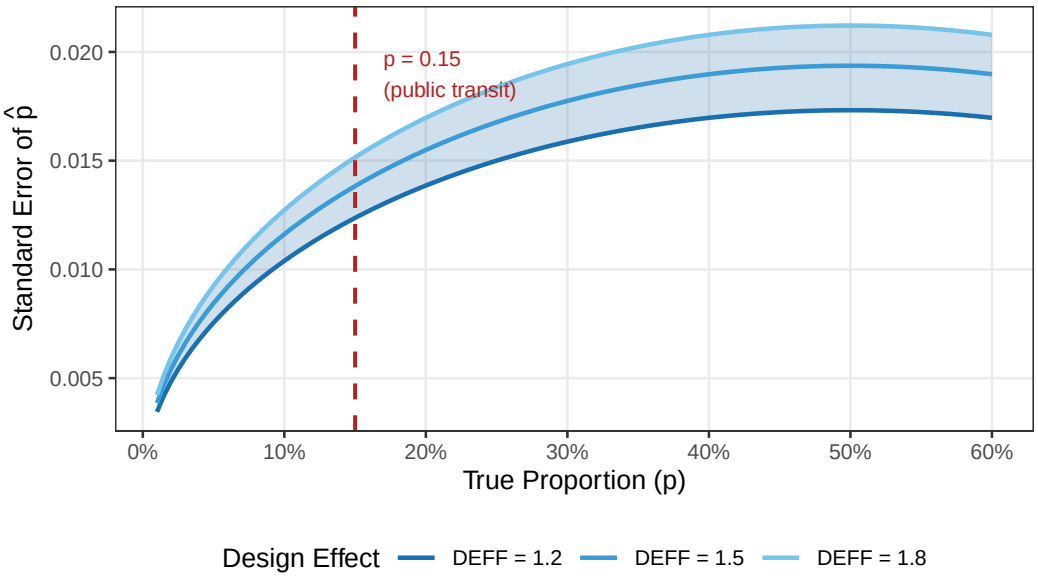

Figure 5: Anticipated Standard Error of an Estimated Proportion by True Proportion and Design Effect (n = 1,000). Dashed vertical line marks p = 0.15, the assumed public-transit proportion. The shaded band spans DEFF = 1.2 to 1.8.

Table 11: Base Weights by Age Group

Age Group	Sampling Fraction	Base Weight
18–34	0.004427	225.9
35–49	0.004905	203.9
50–64	0.004246	235.5
65+	0.004384	228.1

by age group and, where feasible, geographic weighting classes (e.g., groups of sampled tracts with similar response propensities). After nonresponse adjustment, weights should be calibrated to known external population controls—at minimum, age-group-by-sex totals from the 2020 Census. If additional reliable controls are available (e.g., race/ethnicity), raking across multiple marginal distributions may further reduce nonresponse bias and improve the precision of key estimates.

6 Variance Estimation

For design-based variance estimation, analysts should define the selected census tracts as PSUs. Because no explicit stratification is built into the present design, a conservative approach is to treat the entire sample as a single stratum with 40 PSUs. A modestly improved approach is to form variance-estimation strata by pairing adjacent sampled tracts in the sorted PPS frame order, thereby capturing some implicit geographic stratification; this yields 20 strata with 2 PSUs each.
The final analysis data file should contain at minimum: (1) a final respondent weight, (2) a PSU identifier corresponding to the sampled tract FIPS code, and (3) a variance stratum variable. Analysts can then compute Taylor-series linearized standard errors or replicate-weight-based variances using standard survey software (e.g., the `survey` package in R, Stata's `svy` suite, or SAS `PROC SURVEYMEANS`). Because there is only one sampled block group per tract, the tract remains the PSU

for all analyses; treating block groups as PSUs would incorrectly inflate degrees of freedom.

7 Maps

7.1 The City of Detroit

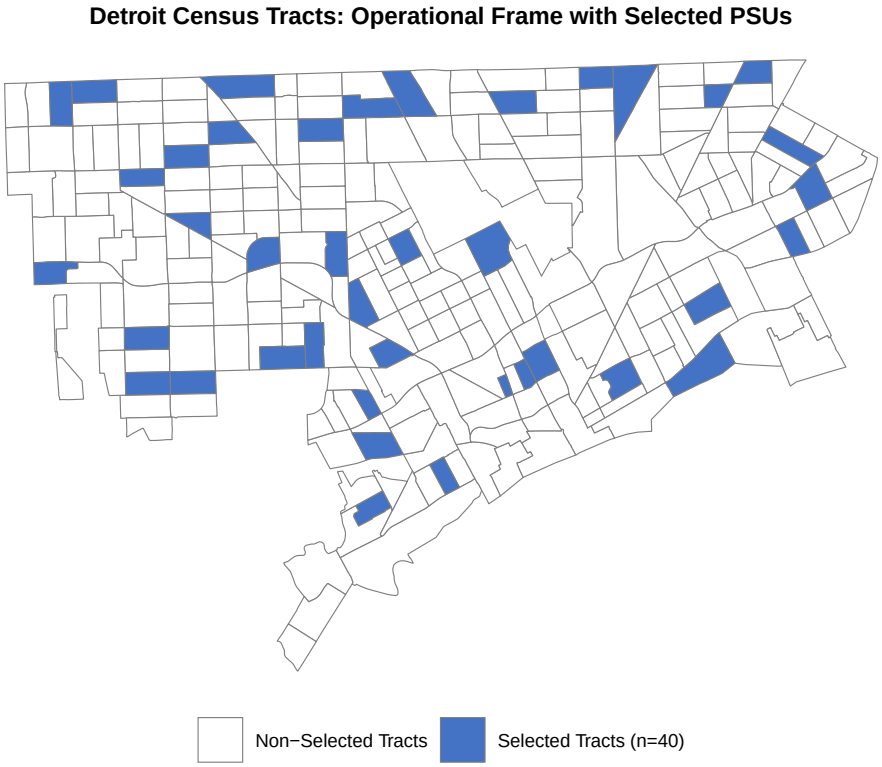

Figure 6: Census Tracts in the Detroit Operational Frame. All 266 tracts meeting the minimum population threshold are shown; the 40 selected PSUs are highlighted in blue.

7.2 Selected Tracts and Block Groups

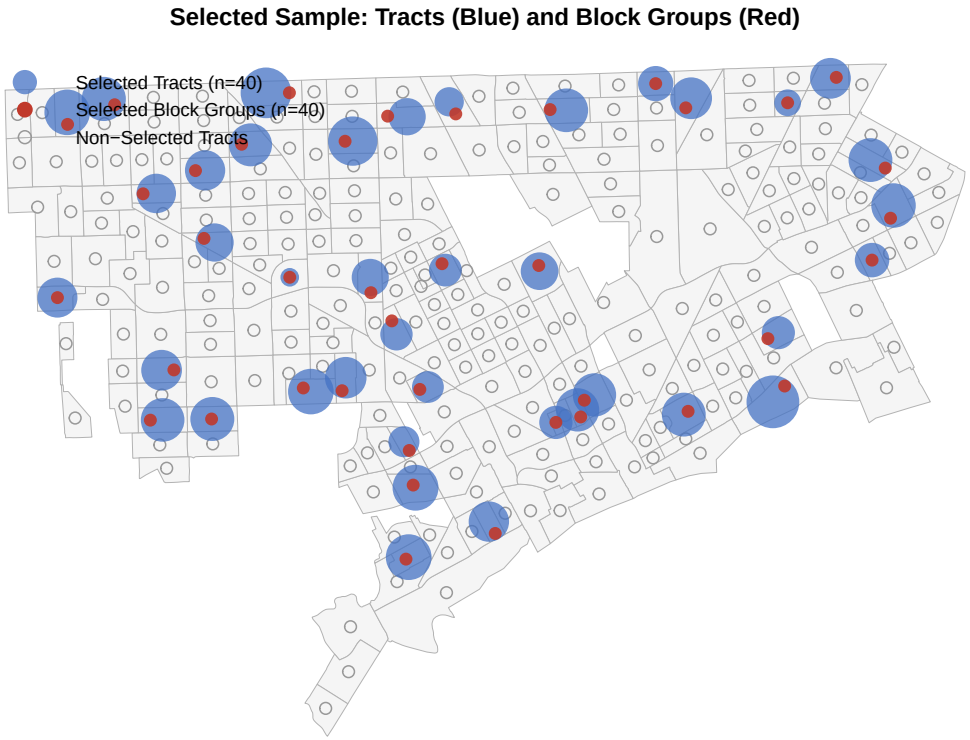

Figure 7: Selected Sample: PSU Tracts (blue circles) and SSU Block Groups (red circles). Circle size is proportional to the composite MOS, illustrating that larger tracts are selected with higher probability under PPS. Each selected tract contains exactly one selected block group (red dot).

8 Appendix

8.1 Codebook of Frame and Sample Files

Table 12: Codebook for the Tract Frame File (FRAME1_NEYMAN.csv)

Variable Name	Description
Tract	Census tract FIPS code
NAME	Tract name/geographic identifier
TotHH	Total households in tract
TotPerson	Total persons in tract
Age18to34	Population aged 18–34
Age35to49	Population aged 35–49
Age50to64	Population aged 50–64
Age65plus	Population aged 65 and older
AgeUnder18	Population under age 18
Adult18plus	Population aged 18 and older
CompMOS	Composite measure of size used for PPS selection
nBGs	Number of block groups in tract

Table 13: Codebook for the Sample File (SAMPLE_NEYMAN.csv)

Variable Name	Description
BlockGroup	Selected block group FIPS code
TractFIPS	Parent tract FIPS code
TotHH	Total households in block group
TotPerson	Total persons in block group
Adult18plus	Population aged 18 and older in block group
Age18to34	Population aged 18–34 in block group
Age35to49	Population aged 35–49 in block group
Age50to64	Population aged 50–64 in block group
Age65plus	Population aged 65+ in block group
CompMOS	Composite measure of size for the block group
pi_tract	First-stage inclusion probability for the tract
pi_bg_given_tract	Conditional second-stage inclusion probability for the block group
pi_overall_bg	Overall block-group selection probability
f_person_18to34	Person sampling rate for age 18–34
f_person_35to49	Person sampling rate for age 35–49
f_person_50to64	Person sampling rate for age 50–64
f_person_65plus	Person sampling rate for age 65+
listed_18to34	Expected number of listed persons selected in age 18–34
listed_35to49	Expected number of listed persons selected in age 35–49
listed_50to64	Expected number of listed persons selected in age 50–64
listed_65plus	Expected number of listed persons selected in age 65+
total_listed	Expected total persons selected before nonresponse
expected_resp_18to34	Expected completed interviews in age 18–34 from fractional rates
expected_resp_35to49	Expected completed interviews in age 35–49 from fractional rates
expected_resp_50to64	Expected completed interviews in age 50–64 from fractional rates
expected_resp_65plus	Expected completed interviews in age 65+ from fractional rates
expected_total_resp	Expected total completed interviews from fractional rates
base_wt_18to34	Base weight for persons aged 18–34
base_wt_35to49	Base weight for persons aged 35–49
base_wt_50to64	Base weight for persons aged 50–64
base_wt_65plus	Base weight for persons aged 65+
select_n_18to34	Operational integer persons to select in age 18–34
select_n_35to49	Operational integer persons to select in age 35–49
select_n_50to64	Operational integer persons to select in age 50–64
select_n_65plus	Operational integer persons to select in age 65+
expected_comp_from_select_18to34	Expected completed interviews from operational 18–34 selections

Table 13: Codebook for the Sample File (SAMPLE_NEYMAN.csv) *(continued)*

Variable Name	Description
expected_comp_from_select_35to49	Expected completed interviews from operational 35–49 selections
expected_comp_from_select_50to64	Expected completed interviews from operational 50–64 selections
expected_comp_from_select_65plus	Expected completed interviews from operational 65+ selections
total_selected	Total operational persons selected in the block group
expected_total_comp_from_select	Expected total completed interviews from operational integer selections

8.2 Listing of Sample PSUs and SSUs with Selection Probabilities and Sampling Rates

Table 14: Selected Tracts and Bl

Tract	BlockGroup	TotHH	TotPerson	Adult18plus	Age18to34	Age35to49	Age50to64	Age65plus	CompMOS	pi_tract
26163500200	261635002002	764	2180	1480	538	394	347	201	6.6688	0.1746
26163500900	261635009002	879	2049	1419	513	390	345	171	6.3985	0.2151
26163501400	261635014002	412	990	709	238	185	189	97	3.1888	0.2241
26163502000	261635020001	766	1858	1289	437	357	346	149	5.8080	0.1078
26163503400	261635034001	494	886	673	203	178	190	102	3.0257	0.0562
26163505100	261635051003	518	1325	1018	310	220	290	198	4.5508	0.1871
26163506100	261635061001	744	1875	1379	507	319	336	217	6.1871	0.1149
26163506900	261635069003	418	905	674	213	177	145	139	3.0362	0.2130
26163508000	261635080001	626	1098	835	270	210	270	85	3.7444	0.0695
26163511400	261635114001	678	1026	832	261	176	206	189	3.7220	0.1388
26163513900	261635139001	453	765	612	169	116	182	145	2.7256	0.0966
26163515700	261635157001	1429	1766	1663	551	278	358	476	7.4097	0.3630
26163516700	261635167003	787	1254	1090	269	226	298	297	4.8667	0.2155
26163517500	261635175002	1205	1392	1282	456	178	324	324	5.6879	0.2080
26163520300	261635203001	575	652	584	291	86	115	92	2.6017	0.2078
26163521800	261635218001	901	1466	1179	282	207	249	441	5.2543	0.0976
26163523300	261635233003	330	934	646	277	165	136	68	2.9112	0.1699
26163524200	261635242001	547	1593	1070	430	347	182	111	4.8650	0.2429
26163525700	261635257004	433	1368	889	298	312	191	88	4.0463	0.2451
26163526400	261635264001	248	742	506	201	144	97	64	2.2886	0.0808
26163530800	261635308002	466	603	444	160	90	112	82	1.9848	0.0906
26163531800	261635318001	465	779	576	191	139	153	93	2.5847	0.0937
26163533600	261635336002	577	816	597	172	169	156	100	2.6911	0.0864
26163534700	261635347001	544	1104	764	237	192	180	155	3.4347	0.1843
26163535600	261635356001	299	659	531	133	114	152	132	2.3720	0.2404
26163536500	261635365002	644	1207	906	275	234	224	173	4.0747	0.1332
26163537200	261635372001	350	583	450	122	131	126	71	2.0289	0.0377
26163538200	261635382001	436	1105	905	193	222	222	268	4.0608	0.1349
26163538600	261635386003	457	1111	882	248	187	251	196	3.9401	0.3010
26163539200	261635392001	1142	2152	1745	396	409	480	460	7.8139	0.3245
26163539600	261635396002	728	1829	1389	391	340	323	335	6.2387	0.2030
26163540400	261635404002	642	1232	962	258	215	286	203	4.3010	0.1684
26163540900	261635409001	452	993	743	212	164	186	181	3.3262	0.2197
26163541500	261635415001	814	1682	1352	373	289	378	312	6.0416	0.2405
26163542300	261635423001	367	858	667	217	161	149	140	2.9968	0.1452
26163543100	261635431002	420	1107	878	206	178	218	276	3.9206	0.1591
26163544000	261635440001	1250	2534	1985	694	473	568	250	8.9001	0.1652
26163545500	261635455001	440	1245	835	308	231	196	100	3.7672	0.2171
26163545900	261635459001	826	1861	1282	494	348	309	131	5.7802	0.2118
26163546700	261635467002	559	1091	823	285	202	210	126	3.6965	0.1741

8.3 Disclosure of AI Use

Two generative AI tools were used to assist with this project:
Claude (Anthropic, claude-sonnet-4-5) was used for: (1) editing and improving the clarity of narrative sections; (2) converting the script from R Markdown to Quarto format and aligning the YAML header with project format requirements; (3) refining table formatting and code organization; and (4) tightening prose to meet the 10–15 page body limit.

ChatGPT (OpenAI, GPT-4o) was used for: (1) brainstorming the structure of the composite MOS formula explanation; (2) drafting an initial version of the weighting plan section; and (3) checking the logic of the self-weighting proof for consistency.
our Team takes responsibility for all methodological decisions, formula derivations, code review, data preparation for this work.
Verbatim prompts used with Claude:

> *"Here is my .qmd script for a multi-stage area probability sample design project. Refine it to match the project requirements. I set echo to false so I could knit it into the report after running the codes. Key things to fix: report formatting (page count, spacing, sections)."*

> *"Review the YAML header and make sure it is correct Quarto syntax, not R Markdown syntax. Add a proper title page separation with newpage, and make sure the Appendix starts on a new page so it clearly falls outside the 10–15 page body limit."*

> *"Tighten the prose in Sections 2–7 to keep the body within the page limit. Consolidate redundant language in the weighting and variance estimation sections."*

Verbatim prompts used with ChatGPT:

> *"I am designing a multi-stage area probability sample for a Detroit transportation survey. I need to explain a composite measure of size that weights age groups by their required sampling fractions. Can you help me write a clear explanation for a technical report audience?"*

> *"Write a weighting plan section for a survey sample design report. The design is self-weighting within each of four age groups (18–34, 35–49, 50–64, 65+), so the base weight for each respondent is the inverse of their age-group sampling fraction. After the base weight, describe nonresponse adjustment and calibration to census controls."*

> *"In a multi-stage PPS design where tracts are PSUs and one block group is selected per tract, the overall person probability equals the product of the tract inclusion probability, the conditional block group probability, and the within-block-group person rate. Show that this simplifies to a constant f_g within each age group, confirming self-weighting."*

8.4 Output Files

Table 15: Output Files Written by the Script

File	Rows
FRAME1_NEYMAN.csv	266
FRAME2_NEYMAN.csv	614
SAMPLE_NEYMAN.csv	40